

TU

جامعة الطائف
Taif University

WiFi Signal Monitoring System

Student Names:

Leen Abdullah Alamri | 44456385

Amna Feras Bettar | 44254743

Bayan Mohammed Alotaibi | 44451677

Rawad Eid Alotaibi | 44450582

Sahar Hassan Althagafi | 44453445

Date:

02 Jun, 2026

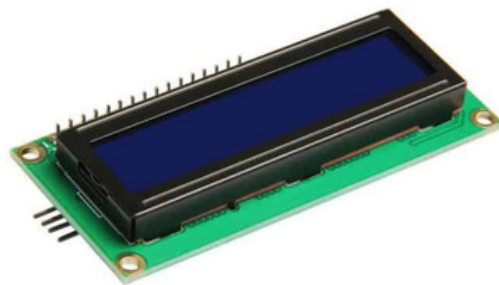
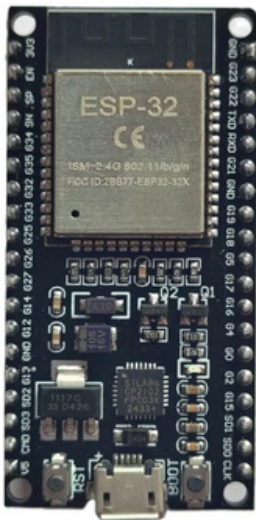


• Introduction

Wireless networks are widely used for communication between devices. However, weak signals or connection failures can affect network performance and data transmission. This project presents a smart system that detects wireless network failures and monitors signal strength using ESP32 microcontrollers.

• Project Objective

The main objective of this project is to monitor the wireless connection between two ESP32 devices and detect any network failure or weak signal condition. The system displays the network status and RSSI value on an LCD screen in real time.



• System Design

The system consists of two ESP32 modules :

- ender ESP32 :

The sender transmits data packets periodically through a wireless connection.

- Receiver ESP32 :

receiver receives the transmitted packets, measures the RSSI (Received Signal Strength Indicator) and displays the network status on the LCD screen.

• Working Principle

- 1/ The sender continuously transmits data packets.
- 2/ The receiver listens for incoming packets.
- 3/ The receiver measures the RSSI value.
- 4/ The LCD displays the network status and signal strength.
- 5/ If communication is lost, the system reports a network failure.

• Network Status Classification

RSSI Value

Network Status

-30 to -60 dBm

Excellent

-61 to -75 dBm

Good

-76 to -85 dBm

Weak

Below -85 dBm

Network Failure Risk

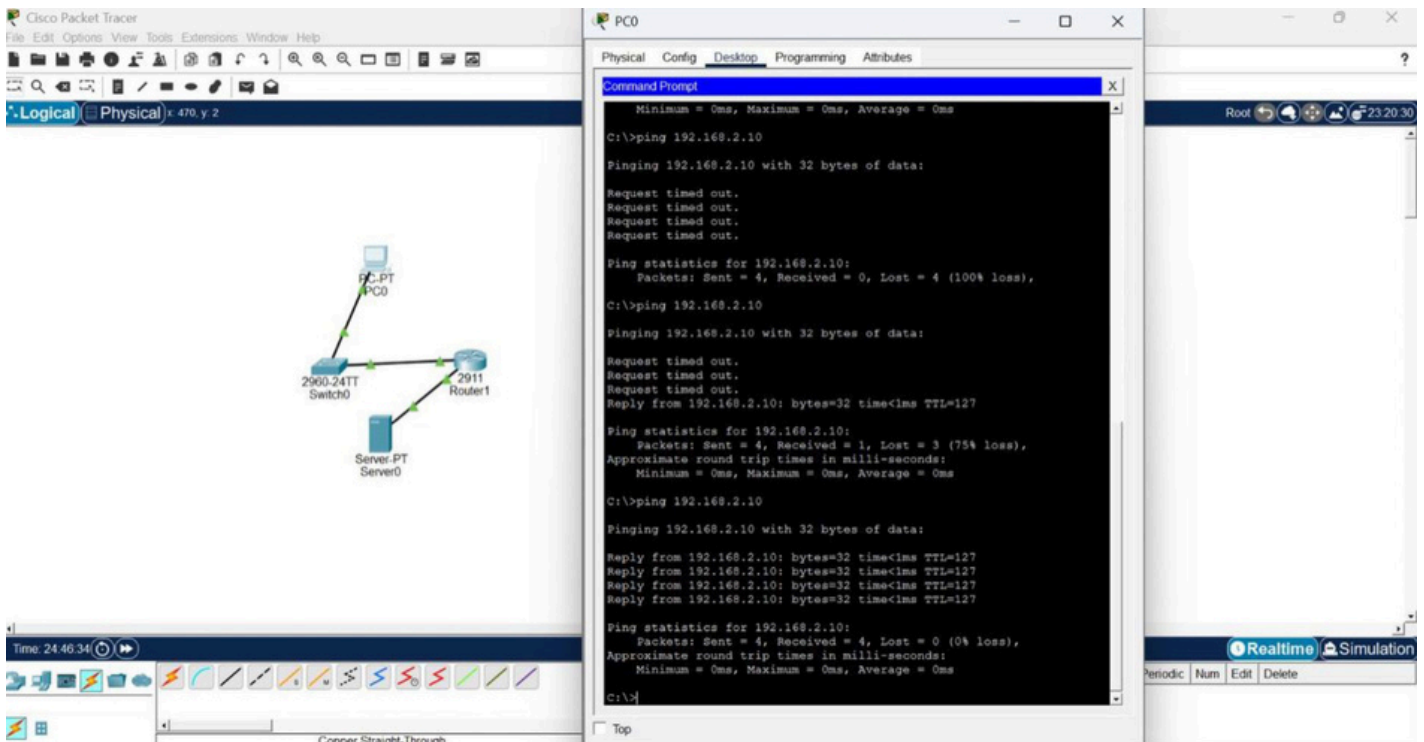
• Results

The system successfully detects wireless network conditions and displays the current status on the LCD screen. Users can monitor signal quality and identify connection problems before a complete network failure occurs.

• Conclusion

This project demonstrates a simple and effective method for wireless network monitoring using ESP32 devices. By displaying RSSI values and connection status in real time, the system helps users detect weak signals and network failures quickly. The project also applies important computer networking concepts such as wireless communication, signal strength monitoring, and network failure detection.

• Results on Packet Tracer

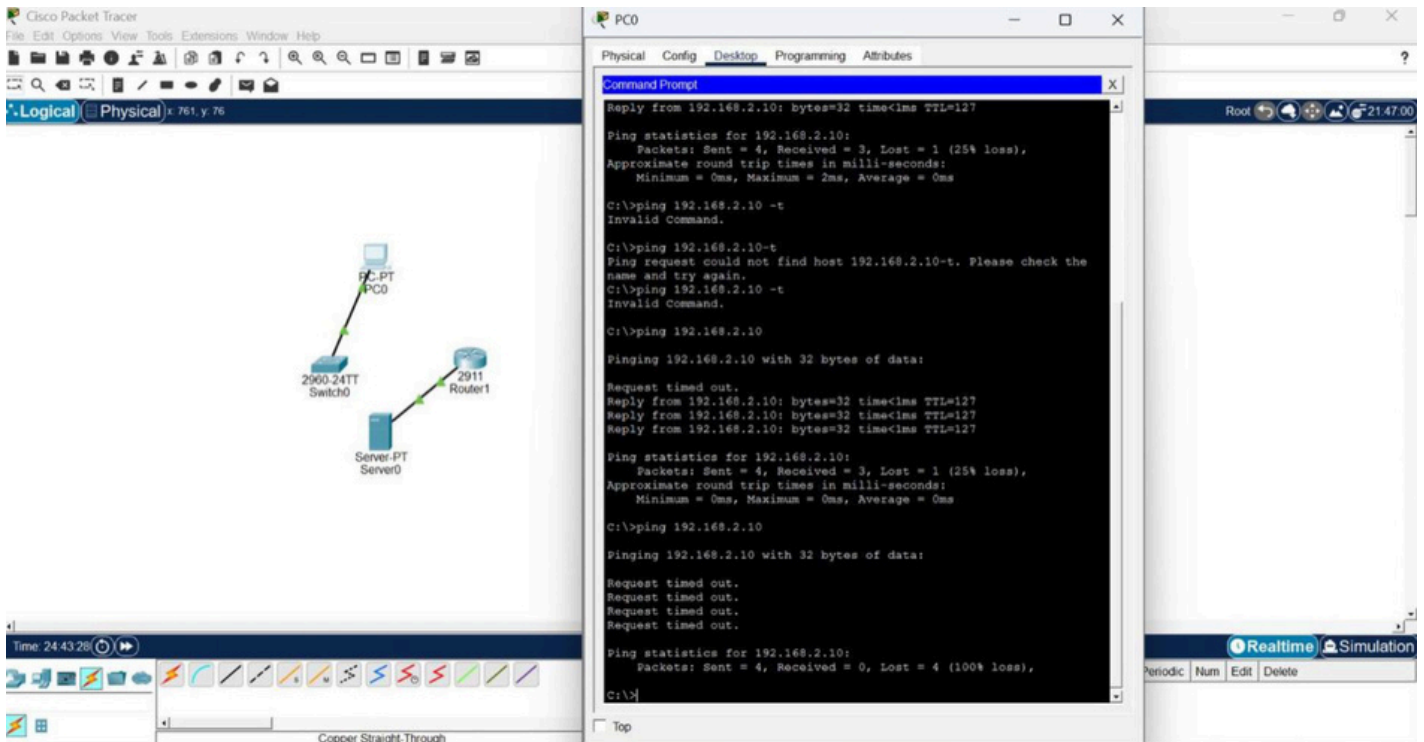


Network Topology & IPs

- PC0: IP: 192.168.1.10 | Gateway: 192.168.1.1
- Server0: IP: 192.168.2.10 | Gateway: 192.168.2.1
- Cables: Connected using Copper Straight-Through cables.

Explaining the Image : Successful Ping Test

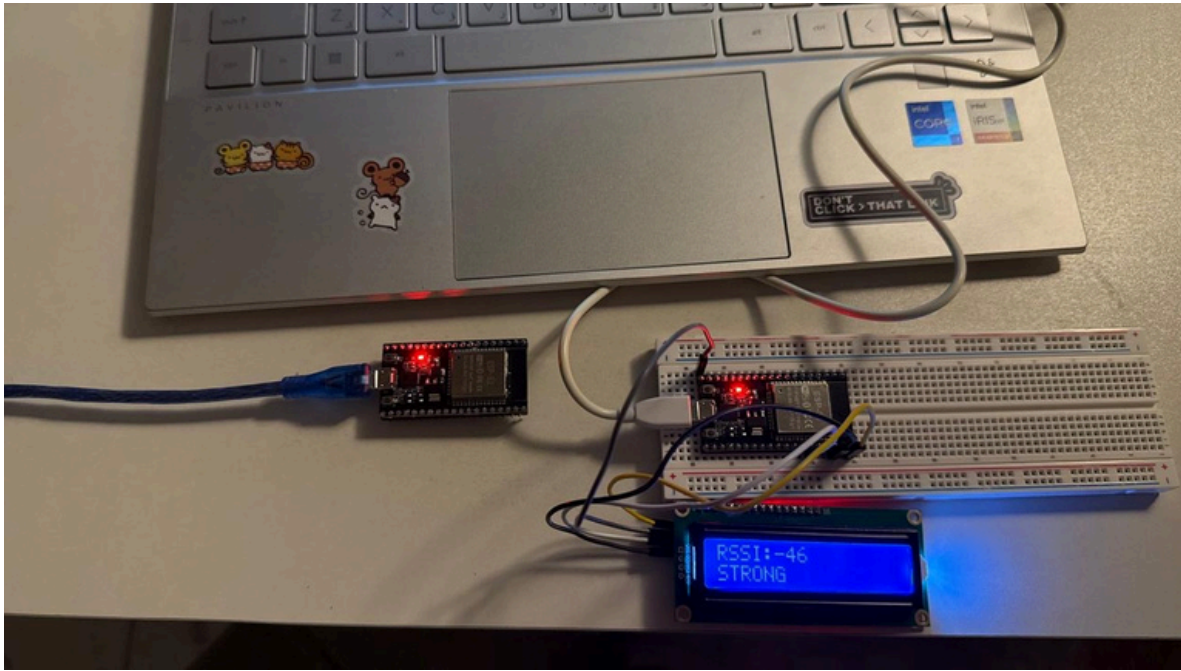
- What happened: I did a ping test from PC0 to Server0 (ping 192.168.2.10).
- First Attempt: It gave "Request timed out" at first because of ARP delay (the devices were discovering each other).
- Final Attempt: The ping was successful with 0% Packet Loss (Sent = 4, Received = 4) and time was < 1 ms. This proves that the network is strong and fully connected.



Network Weakness & Failure Simulation

- What happened: In this step, I simulated a weak/disconnected network by cutting the link or stopping the connection.
- The Result: The ping immediately changed from "Reply" to "Request timed out", resulting in 100 % Packet Loss (Sent = 4, Received = 0).
- Conclusion: This test successfully simulates how network weakness or link failure causes complete data loss between the client and the server.

Implementation and Results hardware



The hardware prototype consists of ESP32 microcontrollers connected through a wireless network. The LCD screen displays the received signal strength indicator (RSSI), allowing users to monitor network quality and detect connectivity degradation.

The codes using Arduino

Tx

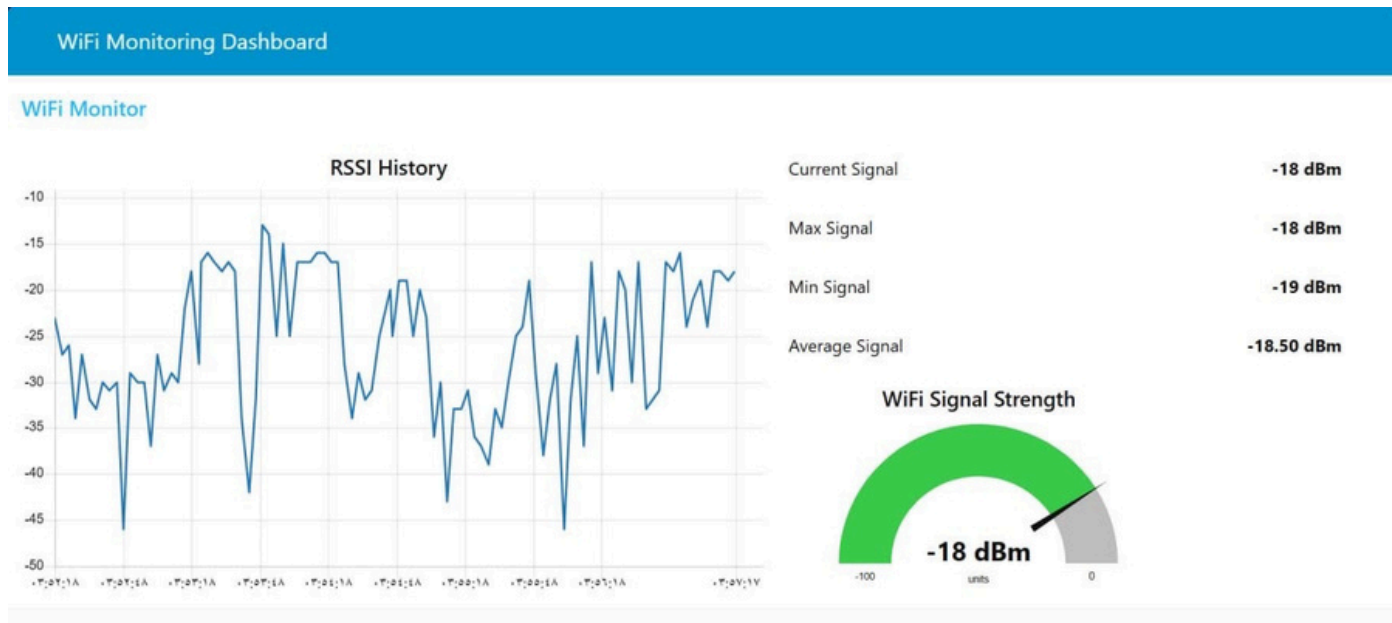
```
NetworksProject\ino  Rows\join
34 Serial.print("IP Address: ");
35 Serial.println(WiFi.localIP());
36 }
37
38 void reconnect() {
39   while (!client.connected()) {
40     Serial.print("Connecting to MQTT... ");
41     if (client.connect("ESP32_Sender", mqtt_user, mqtt_password)) {
42       Serial.println("Connected!");
43     } else {
44       Serial.print("Failed, rc=");
45       Serial.print(client.state());
46       Serial.println();
47       delay(3000);
48     }
49   }
50 }
51
52 Output Serial Monitor X
53 Message (Enter to send message to 'ESP32 Dev Module' on 'COM7')
54
55 RSSI Sent: -36
56 RSSI Sent: -37
57 RSSI Sent: -37
58 RSSI Sent: -42
59 RSSI Sent: -37
60 RSSI Sent: -44
61 RSSI Sent: -39
```

Rx

```
itctch_Jun9aino
67 Serial.println();
68 Serial.println("WiFi Connected");
69 }
70
71 void reconnect() {
72   while (!client.connected()) {
73     Serial.print("Connecting MQTT...");
74     if (client.connect("ESP32_Receiver", mqtt_user, mqtt_password)) {
75       Serial.println("Connected");
76       client.subscribe("signal/rssi");
77     } else {
78       Serial.print("Failed rc=");
79       Serial.println(client.state());
80       delay(3000);
81     }
82   }
83 }
84
85 Input
```

Wi-Fi Monitoring Dashboard

Real-time Wi-Fi monitoring dashboard showing RSSI history and signal strength statistics.



The dashboard provides a graphical representation of Wi-Fi signal strength over time. It displays current, maximum, minimum, and average RSSI values, helping users evaluate network performance and identify potential connectivity issues.